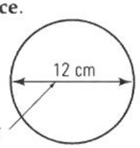
Circumference of a circle and secto

A

The perimeter of a circle is called the **circumference**.
To calculate the circumference:

$C = \pi \times \textit{diameter}$

OR $C = \pi \times d$

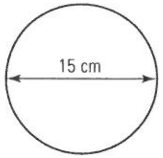


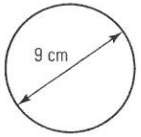
EXAMPLE:

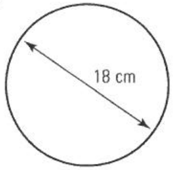
$C = \pi \times 12$

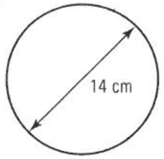
$= 37.7$ cm (one decimal place)

Calculate the circumference of each circle, correct to one decimal place.

1 

2 

3 

4 

5 diameter = 17 cm
6 diameter = 23 cm
7 diameter = 28 cm
8 diameter = 25 cm
9 diameter = 29 cm
10 diameter = 35 cm

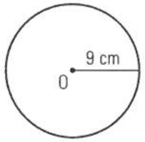
B

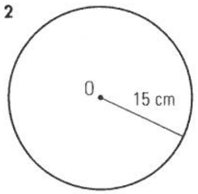
If you are given the radius of a circle, use the rule:

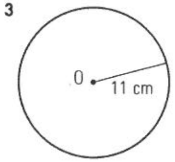
$\textit{Circumference} = 2 \times \pi \times \textit{radius}$

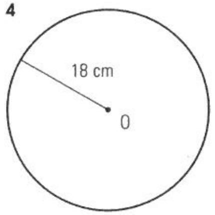
OR $C = 2\pi r$

Calculate the circumference of each circle, correct to one decimal place.
(Note that point O is the centre of each circle.)

1 

2 

3 

4 

5 radius = 6 cm
6 radius = 10 cm
7 radius = 16 cm
8 radius = 19 cm
9 radius = 24 cm
10 radius = 33 cm
11 radius = 1.4 m
12 radius = 2.1 m
13 radius = 2.5 m
14 radius = 2.9 m
15 radius = 3.8 m
16 radius = 3.2 m

Calculate the circumference of each circle, to the nearest millimetre.

17 radius = 45 mm
18 radius = 63 mm
19 diameter = 56 mm
20 diameter = 93 mm
21 radius = 72 mm
22 diameter = 135 mm
23 radius = 59 mm
24 diameter = 86 mm
25 radius = 53 mm

C

A **sector** is a piece of a circle bounded by two straight sides (radii) and a circular arc.
To calculate the **length of the arc** use the size of the angle in the sector (θ).

length of arc $= \frac{\theta}{360} \times 2 \times \pi \times (\text{radius})$

Remember, a **diameter** divides a circle into two equal **semi-circles**.
A **quadrant** is quarter of a circle.

EXAMPLE:

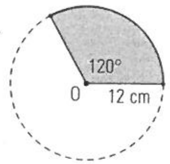
length of arc $= \frac{120}{360} \times 2 \times \pi \times 12$

$= 25.1$ cm (to one decimal place)

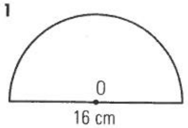
perimeter of sector = length of arc + 2 × radius

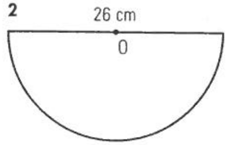
$= 25.1 + 2 \times 12$

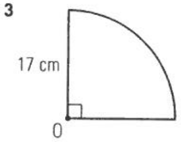
$= 49.1$ cm (to one decimal place)

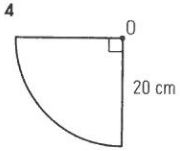


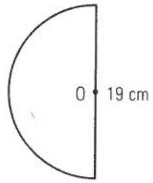
Calculate the perimeter of each sector, correct to one decimal place.

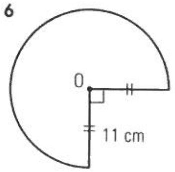
1 

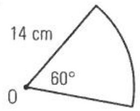
2 

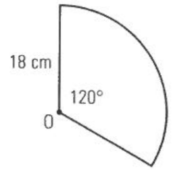
3 

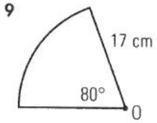
4 

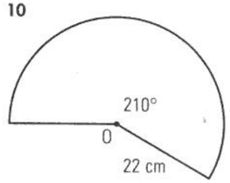
5 

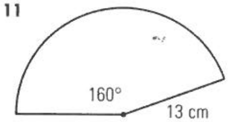
6 

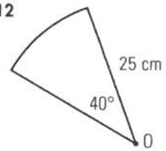
7 

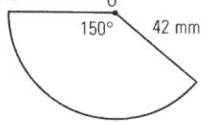
8 

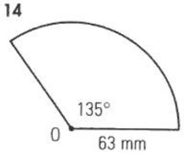
9 

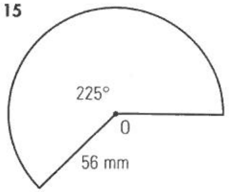
10 

11 

12 

13 

14 

15 

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